Exp 5 - Design of High Pass Filter using Windowing Techniques
5.2. Generate the coefficients of an FIR High-Pass filter using Hamming windowing techniques

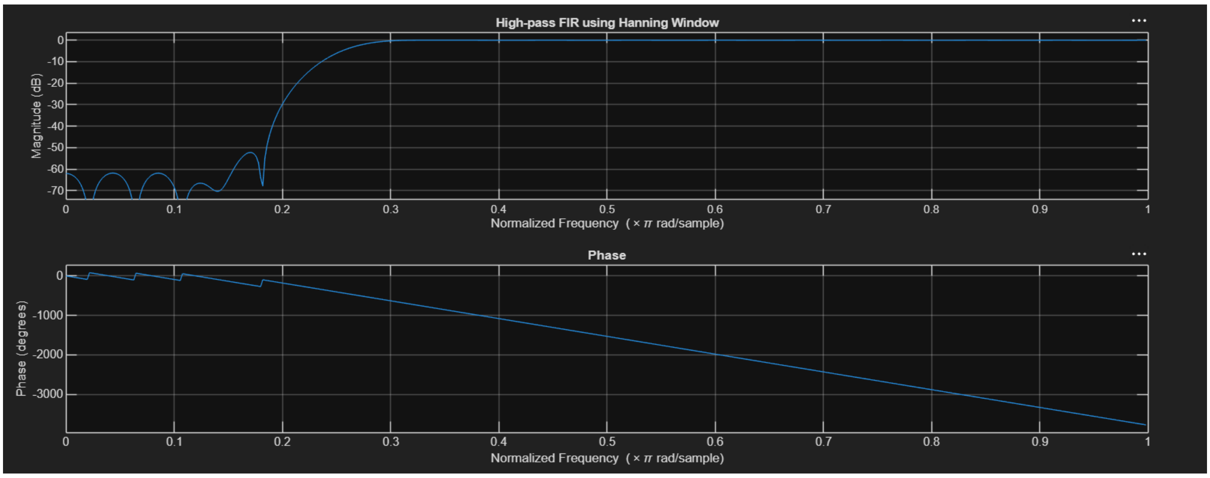
PROGRAM

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000;       % High-pass cutoff frequency (Hz)
fs = 8000;       % Sampling frequency (Hz)
N  = 50;         % Filter order
wc = fp/(fs/2);   % Normalized cutoff (0–1)
%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');
%% --- Frequency Response Plots ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```

OUTPUT

```
--- High-Pass Coefficients (Hamming Window) ---
  -0.0007
        0
   0.0009
   0.0016
   0.0015
        0
  -0.0024
  -0.0044
  -0.0039
        0
   0.0060
   0.0103
```

0.0088
0
-0.0128
-0.0217
-0.0184
0
0.0268
0.0464
0.0410
0
-0.0725
-0.1567
-0.2240
0.7493
-0.2240
-0.1567
-0.0725
0
0.0410
0.0464
0.0268
0
-0.0184
-0.0217
-0.0128
0
0.0088
0.0103
0.0060
0
-0.0039
-0.0044
-0.0024
0
0.0015
0.0016
0.0009
0
-0.0007



High-pass FIR using Hanning Window
Magnitude (dB)
0
-10
-20
-30
-40
-50
-60
-70
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)
Phase
Phase (degrees)
0
-1000
-2000
-3000
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
Normalized Frequency (× π rad/sample)